

DRDO iDEX MALE UAV AERO PISTON ENGINE DIGITAL TWIN

PREDICTIVE HEALTH MANAGEMENT (PHM) & AUTONOMOUS DIAGNOSTIC GCS PLATFORM

Target Platform: TAPAS-BH201 (Rustom-II) | Propulsion: Rotax 914 Turbocharged Aero Engine | Standard: SAE J1939-11

1. Executive Summary & Mission Scope

This platform is a mission-critical **Ground Control Station (GCS) Digital Twin and Prognostic Health Management (PHM) ecosystem** engineered for India's **DRDO iDEX MALE UAV program (TAPAS-BH201)**. Operating at a continuous 1 Hz synchronization cycle, the digital twin fuses real-time telemetry with **first-principles thermodynamic physics models** and **explainable artificial intelligence (XAI)** to transition aero-engine maintenance from reactive overhauls to proactive, condition-based operational readiness.

2. Dual-Engine Hybrid Diagnostic Architecture

The digital twin operates a synchronized dual-path diagnostic pipeline, combining static threshold boundaries with dynamic physics residuals and machine learning prognostics:

Diagnostic Dimension	Methodology & Mathematical Formulation	Operational Impact
Physics & Thermodynamics	ISA standard atmosphere density correction up to 30,000 ft. Generates expected baselines for CHT, EGT, Fuel Flow (L/h), and MAP based on dynamic throttle and RPM curves.	Calculates real-time physics residuals: <i>Residual</i> = <i> Actual - Expected </i> .
AI Anomaly Detection	Unsupervised Isolation Forest algorithm trained on multi-dimensional flight parameters and physics residuals.	Detects early-stage sensor and system drift before conventional alarm thresholds are breached.
Fault Classification (ISP #2)	Multi-Class Gradient Boosted Classifier (XGBoost) identifying 7 core aero-engine failure modes.	Instant classification of Overheating, Misfire, Lubrication Loss, Fuel Restriction, High Vibration.
SHAP Explainability (XAI)	Game-theoretic Shapley Additive Explanations attributing anomaly scores to exact sensor inputs.	Full diagnostic transparency for GCS flight operators and maintenance crews.
Prognostic RUL Regressor	Continuous Remaining Useful Life regression calculating flight hours remaining until mandatory TBO.	Outputs confidence bounds and future 30-minute operational mission feasibility forecasting.

3. Component Life-Cycle Wear & TBO Tracking (7 Subsystems)

The digital twin continuously computes wear degradation percentages and remaining flight hours across seven critical aero-engine subsystems using live operating stresses:

Component ID	Subsystem Name	Max TBO	Primary Wear Acceleration Factors	Rated Status
COMP-IGN-01	Dual Spark Plugs & Coils	200 hrs	CHT thermal stress (>140°C) and ignition misfire occurrences.	Condition-Based
COMP-LUB-02	Oil Filter Element & Pump	100 hrs	Oil pressure delta (<2.5 bar) and scavenge oil temp (>95°C).	Condition-Based
COMP-FUL-03	Fuel Injector Nozzles & Rail	500 hrs	Fuel line flow restrictions and high delivery variance across Cyl 1-4.	Condition-Based
COMP-THM-04	Turbocharger & Wastegate	400 hrs	High Exhaust Gas Temperature (EGT > 780°C) and boost fatigue.	Condition-Based
COMP-COL-05	Coolant Pump Impeller	600 hrs	Coolant thermal cycling and impeller cavitation indexing.	Condition-Based

COMP-ELE-06	Alternator Belt & Power Bus	300 hrs	Electrical current draw load (>42A) and voltage ripple fluctuations.	Condition-Based
COMP-MEC-07	Crankshaft Main Bearings	1000 hrs	Dynamic mechanical vibration FFT harmonics (1X/2X order spikes).	Condition-Based

4. SocketCAN / SAE J1939 Aero-FADEC Protocol Inspector

The digital twin integrates a virtualized hardware-tap SocketCAN interface (can0 / vcan0) running at **500 kbps**. It broadcasts and decodes 29-bit extended CAN identifiers and standard J1939 Parameter Group Numbers (PGNs):

PGN	Broadcast Label	Decoded Engine Parameters	Rate
61444	EEC1 (Electronic Engine Controller 1)	Actual Engine RPM, Engine Demand Torque %, Starter State	100 ms
65262	ET1 (Engine Temperature 1)	Cylinder Head Temp (CHT °C), Exhaust Gas Temp (EGT °C)	1000 ms
65263	EFL_P1 (Engine Fluid Level/Pressure)	Engine Oil Pressure (bar), Fuel Rail Delivery Pressure (bar)	500 ms
65271	VEP1 (Vehicle Electrical Power 1)	Main 28V DC Avionics Bus Voltage, Alternator Current Draw (A)	1000 ms
65253	ENGINE_HOURS (Total Run Time)	Total Accumulated Airframe/Engine Operating Hours (HRS)	1000 ms
65280	PROGN_DIAG (Digital Twin Telemetry)	Composite Health Index %, RUL Flight Hours Remaining	1000 ms

5. Tactical GCS Interface & Flight Crew Features

The Ground Control Station user interface is crafted with a high-durability **Weathered Metallic Steel Plate aesthetic** matching certified military avionics panels:

- **Precision Aviation Meter Cluster:** Uniform circular analog meters for Engine RPM, Manifold Absolute Pressure (MAP), CHT, EGT, Oil Pressure/Temperature, Fuel Flow, Vibration, and Avionics Bus Voltage.
- **Autonomous Maintenance Advisory:** Dynamic action checklist categorizing work orders into *IMMEDIATE (NEXT FLIGHT)*, *URGENT PRE-FLIGHT*, and *ROUTINE 25H/50H/100H* with 1-click technical report generation.
- **Mission History & Multi-Format Log Exporter:** Instant export of flight telemetry into **CSV**, **Excel (.xlsx)**, **JSON**, and **PDF** formats directly from the live dashboard and history consoles.
- **What-If Scenario Simulation:** Dynamic stress test bench supporting live injection of critical failure modes (Misfire, Overheating, Lubrication Loss) and extreme atmospheric profiles.

6. Technology Stack & Deployment Endpoints

Layer	Technology / Framework	Key Role & Protocols
Backend Engine	FastAPI (Python 3.10+) & Uvicorn	REST API & 1 Hz WebSocket stream (/ws/engine).
AI & Analytics	XGBoost, Scikit-Learn, SHAP, NumPy	Dual-engine AI inference, XAI attribution, RUL curves.
Protocol Layer	python-can & SocketCAN (can0)	SAE J1939-11 29-bit CAN bus frame generation & decode.
GCS Frontend	React 18, Vite, Recharts, Lucide	Tactical GCS dashboard with real-time WebSocket state.
Export Engines	jsPDF, XLSX (SheetJS), ReportLab	Multi-format flight telemetry logs & engineering whitepapers.